Leqing Xu

PhD Candidate in Sustainable Energy Technology

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PROFILE

PhD candidate at the University of Nottingham working on sustainable energy technology. Applies computer science and machine learning to photovoltaic systems, with a focus on very short-term solar forecasting. Training spans electrical and computer engineering, energy systems, and automation.

EDUCATION

PhD in Sustainable Energy Technology 2023 – Present
University of Nottingham, United Kingdom
Focus: short-term solar forecasting using machine learning

M.S. in Electrical and Computer Engineering 2021 – 2022
Purdue University, West Lafayette, United States
Focus: natural language processing and machine learning

M.S. in Energy Systems 2020 – 2021
Northeastern University, United States
Focus: design of carbon materials for water desalination

B.Eng. in Automation 2015 – 2019
Jiangnan University, China
Foundation in control systems and automated processes

RESEARCH INTERESTS

Deep learning for solar energy forecasting; renewable energy systems optimization; physically interpretable very short-term irradiance forecasting.

PUBLICATIONS

Leqing Xu, Liwenbo Zhang, Yuexing Yang, Mengna Zheng, and Yupeng Wu. "From reactive to proactive: A physically interpretable multi-site framework for very short-term solar forecasting." *Energy*, vol. 360, art. no. 141488, 2026. https://doi.org/10.1016/j.energy.2026.141488

Fangqi Chen, **Leqing Xu**, Yanpei Tian, Andrew Caratenuto, Xiaojie Liu, and Yi Zheng. "Electrospun polycaprolactone nanofiber composites with embedded carbon nanotubes/nanoparticles for photothermal absorption." *ACS Applied Nano Materials*, vol. 4, no. 5, pp. 5230–5239, 2021. https://doi.org/10.1021/acsanm.1c00623

PATENT

"An Intelligent Anti-theft Socket System." China National Intellectual Property Administration. Patent no. CN207782050U.